4-Router Ring Topology

Cisco Packet Tracer Step-by-Step Lab Guide

■ Lab Overview

This lab builds a 4-router ring topology with static routing. Each router connects to 2 switches, each switch connects to 2 PCs, and all routers are interconnected via serial links.

Component	Device	Quantity
Routers	Cisco 1941	4 (R1–R4)
Switches	2960-24TT	8 (SW1–SW8)
PCs	PC-PT	16 (PC1–PC16)
Serial Module	HWIC-2T	2 per router

STEP 1 — Add HWIC-2T Modules to All Routers

Each 1941 router needs 2x HWIC-2T modules to get serial ports Se0/0/0 and Se0/1/0.

Do this for R1, R2, R3, and R4:

- Click the router → Physical tab
- Power OFF the router (click power button until light goes off)
- Drag HWIC-2T into Slot 0
- Drag another HWIC-2T into Slot 1
- Power the router back ON → click OK

■ After adding modules you should see: Se0/0/0, Se0/0/1, Se0/1/0, Se0/1/1

STEP 2 — Connect All Devices

Router-to-Router (Serial DCE/DTE Cable)

From	Port	To	Port
R1	Se0/0/0	R2	Se0/0/0
R2	Se0/1/0	R3	Se0/0/0
R3	Se0/1/0	R4	Se0/0/0
R4	Se0/1/0	R1	Se0/1/0

Switches to Routers (Copper Straight-Through)

Switch	Router	Router Port
SW1	R1	Giga0/0
SW2	R1	Giga0/1
SW3	R2	Giga0/0
SW4	R2	Giga0/1
SW5	R3	Giga0/0
SW6	R3	Giga0/1
SW7	R4	Giga0/0
SW8	R4	Giga0/1

PCs to Switches (Copper Straight-Through)

PC	Switch	PC	Switch
PC1, PC2	SW1	PC9, PC10	SW5
PC3, PC4	SW2	PC11, PC12	SW6
PC5, PC6	SW3	PC13, PC14	SW7
PC7, PC8	SW4	PC15, PC16	SW8

STEP 3 — Configure Routers (CLI)

■ Replace 'last_name' in the banner with your actual last name.

Router R1

```
enable
configure terminal
hostname R1
line console 0
password tacocat1
login
exit
enable secret redhot1
banner motd # Now accessing R1. Authorized Personnel only - Admin last_name #
clock set 18:00:00 27 November 2017
interface giga0/0
ip address 192.168.10.1 255.255.255.0
no shutdown
exit
interface giga0/1
ip address 192.168.11.1 255.255.255.0
no shutdown
exit
interface se0/0/0
ip address 11.0.0.1 255.255.255.252
```

```
clock rate 64000
no shutdown
exit
interface se0/1/0
ip address 14.0.0.2 255.255.255.252
no shutdown
exit
end
write memory
```

Router R2

```
enable
configure terminal
hostname R2
line console 0
password tacocat2
login
exit
enable secret redhot2
banner motd # Now accessing R2. Authorized Personnel only - Admin last_name #
clock set 18:00:00 27 November 2017
interface giga0/0
ip address 10.1.1.1 255.255.255.0
no shutdown
exit
interface giga0/1
ip address 10.1.2.1 255.255.255.0
no shutdown
exit
interface se0/0/0
ip address 11.0.0.2 255.255.255.252
no shutdown
exit
interface se0/1/0
ip address 12.0.0.1 255.255.255.252
clock rate 64000
no shutdown
exit
end
write memory
```

Router R3

```
enable
configure terminal
hostname R3
line console 0
password tacocat3
login
exit
enable secret redhot3
banner motd # Now accessing R3. Authorized Personnel only - Admin last_name #
clock set 18:00:00 27 November 2017
```

```
interface giga0/0
ip address 172.16.22.1 255.255.255.0
no shutdown
exit
interface giga0/1
ip address 172.16.23.1 255.255.255.0
no shutdown
exit
interface se0/0/0
ip address 12.0.0.2 255.255.255.252
no shutdown
exit
interface se0/1/0
ip address 13.0.0.1 255.255.255.252
clock rate 64000
no shutdown
exit
end
write memory
```

Router R4

```
enable
configure terminal
hostname R4
line console 0
password tacocat4
login
exit
enable secret redhot4
banner motd # Now accessing R4. Authorized Personnel only - Admin last_name #
clock set 18:00:00 27 November 2017
interface giga0/0
ip address 192.0.41.1 255.255.255.0
no shutdown
exit
interface giga0/1
ip address 192.0.42.1 255.255.255.0
no shutdown
exit
interface se0/0/0
ip address 13.0.0.2 255.255.255.252
no shutdown
exit
interface se0/1/0
ip address 14.0.0.1 255.255.255.252
no shutdown
exit
end
write memory
```

STEP 4 — Add Static Routes

■ Each router must know routes to ALL other networks.

R1 Static Routes

```
configure terminal
ip route 10.1.1.0 255.255.255.0 11.0.0.2
ip route 10.1.2.0 255.255.255.0 11.0.0.2
ip route 172.16.22.0 255.255.255.0 11.0.0.2
ip route 172.16.23.0 255.255.255.0 11.0.0.2
ip route 192.0.41.0 255.255.255.0 14.0.0.1
ip route 192.0.42.0 255.255.255.0 14.0.0.1
ip route 12.0.0.0 255.255.255.252 11.0.0.2
ip route 13.0.0.0 255.255.255.252 14.0.0.1
end
write memory
```

R2 Static Routes

```
configure terminal
ip route 192.168.10.0 255.255.255.0 11.0.0.1
ip route 192.168.11.0 255.255.255.0 11.0.0.1
ip route 172.16.22.0 255.255.255.0 12.0.0.2
ip route 172.16.23.0 255.255.255.0 12.0.0.2
ip route 192.0.41.0 255.255.255.0 12.0.0.2
ip route 192.0.42.0 255.255.255.0 12.0.0.2
ip route 13.0.0.0 255.255.255.252 12.0.0.2
ip route 14.0.0.0 255.255.255.252 11.0.0.1
end
write memory
```

R3 Static Routes

```
configure terminal
ip route 192.168.10.0 255.255.255.0 12.0.0.1
ip route 192.168.11.0 255.255.255.0 12.0.0.1
ip route 10.1.1.0 255.255.255.0 12.0.0.1
ip route 10.1.2.0 255.255.255.0 12.0.0.1
ip route 192.0.41.0 255.255.255.0 13.0.0.2
ip route 192.0.42.0 255.255.255.0 13.0.0.2
ip route 11.0.0.0 255.255.255.252 12.0.0.1
ip route 14.0.0.0 255.255.255.252 13.0.0.2
end
write memory
```

R4 Static Routes

```
configure terminal
ip route 192.168.10.0 255.255.255.0 14.0.0.2
ip route 192.168.11.0 255.255.255.0 14.0.0.2
ip route 10.1.1.0 255.255.255.0 13.0.0.1
ip route 10.1.2.0 255.255.255.0 13.0.0.1
ip route 172.16.22.0 255.255.255.0 13.0.0.1
ip route 172.16.23.0 255.255.255.0 13.0.0.1
ip route 11.0.0.0 255.255.255.252 14.0.0.2
ip route 12.0.0.0 255.255.255.252 13.0.0.1
end
```

```
write memory
```

STEP 5 — Assign IP Addresses to PCs

Click each PC → Desktop tab → IP Configuration and enter the following:

PC	IP Address	Subnet Mask	Default Gateway
PC1	192.168.10.2	255.255.255.0	192.168.10.1
PC2	192.168.10.3	255.255.255.0	192.168.10.1
PC3	192.168.11.2	255.255.255.0	192.168.11.1
PC4	192.168.11.3	255.255.255.0	192.168.11.1
PC5	10.1.1.2	255.255.255.0	10.1.1.1
PC6	10.1.1.3	255.255.255.0	10.1.1.1
PC7	10.1.2.2	255.255.255.0	10.1.2.1
PC8	10.1.2.3	255.255.255.0	10.1.2.1
PC9	172.16.22.2	255.255.255.0	172.16.22.1
PC10	172.16.22.3	255.255.255.0	172.16.22.1
PC11	172.16.23.2	255.255.255.0	172.16.23.1
PC12	172.16.23.3	255.255.255.0	172.16.23.1
PC13	192.0.41.2	255.255.255.0	192.0.41.1
PC14	192.0.41.3	255.255.255.0	192.0.41.1
PC15	192.0.42.2	255.255.255.0	192.0.42.1
PC16	192.0.42.3	255.255.255.0	192.0.42.1

STEP 6 — Test Connectivity

Go to any PC → Desktop → Command Prompt and run these pings:

```
ping 192.168.10.1    (your own gateway - should always work)
ping 192.168.11.2    (same router, different subnet)
ping 10.1.1.2        (different router)
ping 192.0.42.3      (farthest PC)
```

■ The first ping may lose 1-2 packets due to ARP — this is normal!

■ Quick Reference — Troubleshooting

Problem	Command	Fix
Serial link red	show controllers se0/X/X	Add clock rate on DCE side
Interface down	show ip interface brief	Run no shutdown on interface

Ping fails	show ip route	Check static routes are present
Wrong port	show run	Verify IP on correct interface